

PHARMACOEPIDEMIOLOGY, CHAPTER 38

Studies of Medication Adherence

A Plain-Language Guide

Why patients don't always take medications as prescribed, how researchers try to measure that gap, and what the numbers can and cannot tell us.

Prepared by Omokhodion Alfred Eriakha

Based on Lauffenburger, Acri, and Gross, “Studies of Medication Adherence,” in Pharmacoepidemiology, 6th Edition

What Is Medication Adherence?

Adherence is the extent to which a patient's medication-taking behavior matches what was agreed with their health care provider. The word choice matters: the older term “compliance” pictured a patient passively obeying doctor's orders, while “adherence” treats the patient as an active partner in the treatment plan.

Researchers break adherence into three distinct phases, covered next: initiation, implementation, and persistence. A patient can succeed at one phase and struggle with another.

Three Phases, One Timeline

Adherence unfolds over time rather than happening all at once. The chapter's framework (its Figure 38.1) breaks a patient's course on a medication into three sequential phases, each measured in its own way.

1. Initiation

Taking the first dose of a newly prescribed medication.

2. Implementation

How closely actual dosing matches the prescribed regimen over time.

3. Persistence

How long the patient keeps taking it before stopping altogether.

The next three slides look at each phase in turn.

Initiation: The First Dose

Initiation is the moment a patient takes the first dose of a medication that has just been prescribed. It sounds like a formality, but it is not guaranteed: a prescription can be written and never acted on at all.

When a new prescription is never filled or never started, that gap is called primary nonadherence. It is surprisingly hard to detect from pharmacy records alone, since a prescription that was never filled leaves no dispensing record to find — a puzzle this deck returns to later.

Implementation: The Day-to-Day

Implementation is how closely a patient's actual dosing — timing, amount, and frequency — matches the prescribed regimen across the whole time they are on therapy.

This is the phase most measurement methods are built to capture, usually summarized as a percentage over a chosen stretch of time — for example, the share of prescribed doses actually taken during a three-month period. Nearly every method covered in this deck is, at heart, a way of estimating implementation.

Persistence: Staying on Therapy

Persistence is the length of time from initiation until the patient stops taking the medication altogether — the point called discontinuation, whether that happens on a doctor's advice or entirely on the patient's own.

Implementation and persistence are related but distinct. A patient can take every dose correctly for two months and then stop (good implementation, poor persistence), or stay on a drug for years while frequently missing doses (poor implementation, good persistence). Both dimensions matter for outcomes, and neither one alone tells the whole story.

WHY IT MATTERS, 1 OF 7

“Drugs Don't Work in Patients Who Don't Take Them”

The line is attributed to former US Surgeon General C. Everett Koop, and it frames this whole chapter: even the most rigorously tested, effective medication provides zero benefit if it is never actually taken.

Up to 50%

of patients don't take medicines as prescribed

\$100 Billion+

in excess US health spending each year

About 11%

of hospitalizations linked to nonadherence

Up to 30%

of new prescriptions may never be filled

Nonadherence's consequences ripple outward — into the barriers patients face day to day, into how trial results should be read, and out to public health, covered next.

Why Trials Need to Know

Nonadherence inside a clinical trial can dilute a drug's true effect toward the null: if participants in both arms take the study medication inconsistently, a real benefit can be masked simply because the drug was not given a fair chance to work.

That is part of why a trial's measured efficacy — the effect seen under close, supportive monitoring — often runs ahead of a drug's real-world effectiveness, once it is prescribed under ordinary conditions where adherence tends to be lower.

Adherence also changes how safety gets read: no exposure means no chance to see the drug's toxicity, and a patient's adherence quietly dropping can itself be an early sign that side effects are driving them away from treatment.

A Taxonomy of Barriers

Nonadherence rarely has one single cause. The chapter groups its barriers into three broad categories — patient-specific, medication-specific, and health-system-specific. This deck spreads those into five cards for clarity; each still belongs to one of the chapter's three groups.

Patient-Related

Forgetfulness, beliefs about the medication, health literacy.

Medication-Related

Side effects, dosing complexity, cost of the drug itself.

Condition-Related

Chronic, asymptomatic conditions are easy to deprioritize.

Health-System-Related

Poor communication, limited access, fragmented follow-up.

Socioeconomic

Cost, insurance gaps, transportation to pharmacy or clinic.

WHY IT MATTERS, 4 OF 7

Regimen Complexity and Pill Fatigue

As the number of pills and the number of doses per day climb, adherence tends to fall. A once-daily, single-pill regimen is simply easier to sustain, month after month, than one that demands several drugs at several different times a day.

Pill fatigue is a related but distinct problem: even a simple regimen can wear on a patient emotionally over months or years, separate from how complex it is. Both sit inside a broader distinction: nonadherence can be unintentional — forgetting, confusion — or intentional, a deliberate, dose-by-dose decision to skip a medication, often because side effects make it inconvenient or a patient doubts it is still needed. A medication stock-out, by contrast, is a system barrier, not a patient's choice.

The Risk of Taking Too Much

Nonadherence is not only about taking too little. It also includes taking extra or duplicate doses — for instance, after forgetting whether a dose was already taken, or deliberately doubling up to “catch up” on a missed one.

Overuse carries its own risk of toxicity and adverse events. A complete picture of adherence has to account for deviation in both directions, not just doses that were missed.

There's No Universal Threshold

No single percentage of doses taken reliably separates “adherent” from “nonadherent” across every drug and condition. The right cutoff depends on how forgiving the drug's pharmacology is, the condition being treated, and the outcome that matters.

An 80% threshold shows up constantly in the literature as a convention, but treating it as a universal rule oversimplifies a relationship that genuinely varies by drug and disease — a debate this deck comes back to later.

Public Health Stakes: TB and HIV

For infections like tuberculosis and HIV, incomplete adherence is not just an individual risk. A partially treated infection can develop drug resistance — and a resistant strain can then spread to other people entirely.

That is part of why some tuberculosis programs use Directly Observed Therapy, where a health worker watches each dose being swallowed. For these diseases, adherence is treated as a public health matter, not only a personal one.

Why Measurement Is Hard

Given how much adherence matters, it would help to measure it simply and accurately. Neither turns out to be true: every available method is an imperfect proxy for what a patient actually did, and each has its own built-in bias.

This section covers the two biggest traps — the Hawthorne effect and the healthy-adherer effect — before the deck turns to the specific measurement methods themselves.

MEASUREMENT PROBLEM, 2 OF 6

The Gold Standard Is Rarely Practical

Directly observed therapy — a health worker or researcher watching each dose actually go down — comes closest to a genuine gold standard for measuring adherence.

In practice, though, it is rarely usable at scale: it is burdensome to run beyond small or high-stakes programs, and the very act of being watched can change the behavior being measured. Every other method in this deck is, in effect, a workaround for that basic tension.

MEASUREMENT PROBLEM, 3 OF 6

The Hawthorne Effect

People tend to change their behavior when they know they are being watched. In adherence research, this is called the Hawthorne effect, and it shows up constantly.

A patient enrolled in a study that tracks their pill-taking — especially with a visible bottle cap monitor or a diary to fill in — often takes medication more faithfully than they otherwise would. That can make measured adherence look better than real-world adherence actually is, once the study ends and the watching stops.

MEASUREMENT PROBLEM, 4 OF 6

The Healthy-Adherer Effect

Patients who reliably take their medication also tend to do other health-promoting things — exercise more, eat differently, keep preventive appointments, smoke less. Those habits improve outcomes on their own, with nothing to do with the drug itself.

This is a genuine confounding problem for observational adherence-outcome studies: good adherence can look protective even when the pill being adhered to does nothing at all, because the kind of person who adheres well differs systematically from the kind of person who does not.

MEASUREMENT PROBLEM, 5 OF 6

The Propranolol Trial, Proven

A randomized trial of propranolol after heart attack gave the healthy-adherer effect a clean demonstration. Among patients actually taking propranolol, poor adherers had roughly 2.8 times the odds of death compared with good adherers.

The catch: among patients randomized to placebo, poor adherers had almost the same disadvantage — roughly 2.7 times the odds of death compared with good placebo-adherers. Since a placebo has no pharmacological effect, that difference could not be coming from the drug. It came from whatever kind of person a good adherer tends to be.

MEASUREMENT PROBLEM, 6 OF 6

Interval and Granularity Issues

Beyond bias, measurement is also shaped by two choices researchers have to make: what time window to measure over (the interval), and what unit to count in — a single dose, a full day, or a pharmacy refill (the granularity).

The same underlying behavior can look different depending on how the time period is sliced and what unit it is counted in. This deck comes back to both choices once the individual measurement methods have been covered.

A related trap: adherence tends to look higher among prevalent users, since early discontinuers have often already dropped out of that group by the time it is measured. Many studies focus on incident (new) users to avoid this — though a prevalent-user study can still be the more relevant one, since new starters are usually a small share of everyone currently on a drug.

SELF-REPORTS, 1 OF 3

Just Ask the Patient

The simplest, cheapest, and most common measurement approach is also the most direct: ask the patient, or their caregiver, how well they have been taking their medication.

This section is the first of seven measurement-method sections. Each covers how a method works, plus what it captures well and what it misses.

SELF-REPORTS, 2 OF 3

A Few Common Tools

Rather than an open-ended question, most studies use a validated instrument, so results can be compared across patients and across studies.

MMAS-8

The 8-item Morisky Medication Adherence Scale, widely used across chronic conditions.

ACTG Questionnaire

Developed by the AIDS Clinical Trials Group; a mainstay of HIV adherence research.

Visual Analog Scale

The patient marks a point on a line from zero to 100% of doses taken.

SELF-REPORTS, 3 OF 3

Strengths and Limits

Strengths

Cheap and fast to administer at scale, and the only method that can ask a patient why they missed a dose.

Limits

Prone to recall bias and to social desirability bias — patients tend to round their adherence upward for their clinician or interviewer.

Self-report tends to have high specificity but low sensitivity for nonadherence: patients over-report far more often than they under-report. That gives a useful rule of thumb — an admission of nonadherence is usually credible, while a claim of perfect adherence is less reassuring. Self-reported adherence also tends to correlate less strongly with outcomes than electronic monitoring or dispensing data do.

Reading Refill Records

If a patient does not refill a prescription on schedule, that is a signal of possible nonadherence — and pharmacy claims already record every fill date and days-supply, for free, as a byproduct of dispensing the drug.

That is why dispensing data is the workhorse of large database studies: no extra data collection is needed, and it scales to entire populations. Two metrics dominate the field — PDC and MPR, covered next.

Proportion of Days Covered (PDC)

PDC is the number of days in a defined period during which the patient had medication on hand, based on fill dates and days-supply, divided by the total number of days in that period.

PDC is especially useful when a patient is on several medications at once, because it is built so that it can never exceed 100% — overlapping fills are simply not double-counted, unlike the alternative metric covered next.

Medication Possession Ratio (MPR)

MPR is the total days-supply of every fill during a period, divided by the number of days in that period. It uses the same underlying data as PDC — fill dates and days-supply — but adds it up differently.

Because MPR simply sums days-supply across fills, it can exceed 100% when a patient refills early or stockpiles medication — a mathematical quirk that is the key practical difference from PDC.

PDC Versus MPR, Side by Side

PDC

Caps at 100%. Useful for combining adherence across several concurrent drugs without double-counting overlap.

MPR

Can exceed 100%. Simpler to compute for a single drug. Still common, especially in older studies.

Which one a study uses often comes down to researcher preference, database structure, and the drug's pharmacology — in practice, the two tend to produce quite similar estimates.

The Gap Rule and Sensitivity Analysis

Neither PDC nor MPR is fully automatic. Researchers still have to decide how many days without medication on hand counts as a meaningful gap, or how an early refill should be treated — the so-called gap rule.

These seemingly small definitional choices can meaningfully change a study's results, which is why sensitivity analyses that vary the gap rule are recommended practice rather than an optional extra.

Strengths and Limits

Strengths

Objective, scalable to entire populations, and already sitting in databases like the encounter databases covered elsewhere in this series.

Limits

A filled prescription only proves possession, not ingestion — and records are incomplete if a patient uses more than one pharmacy or payer.

In short: dispensing data measures whether the medication left the pharmacy, not whether it was swallowed. Three specific traps compound that gap: automatic-refill programs can make adherence look better than it is; a refill gap cannot distinguish a clinician telling a patient to stop from the patient quietly stopping on their own; and a short measurement window can make a brand-new prescription's first supply look like automatic 100% adherence, before there has been any real chance to lapse.

Counting What's Left

A researcher counts the pills remaining in a bottle at a follow-up visit and compares that count to how many should be left if the patient took every dose as prescribed.

Counts can be announced, where the patient knows in advance and brings the bottle in, or unannounced, where a researcher visits unexpectedly. Unannounced counts guard against a specific trick covered next.

Medication Diaries

Patients log each dose as they take it, on paper or electronically. Done faithfully, a diary can capture timing with real precision — far more than a single follow-up pill count ever could.

The catch is built into the method: a diary is still a real-time self-report, so it inherits self-report's weaknesses. Entries can be forgotten and filled in from memory just before a clinic visit, right when accuracy matters most.

Strengths and Limits

Strengths

Low-cost, concrete, and need no electronic infrastructure to carry out.

Limits

Easily gamed — patients can discard extra pills before a visit — and neither method confirms a pill was actually swallowed.

A pill count only shows that pills are gone from the bottle, and a diary entry only shows that a patient wrote something down — neither is direct proof of what happened in between.

Electronic Drug Monitoring

A pill bottle cap or blister pack fitted with a small chip records the exact date and time it was opened, giving a timestamped log of dosing behavior without relying on the patient to report anything.

It is not foolproof: a patient can open the container without taking the dose, or empty several days' pills into a separate weekly pillbox, defeating the monitor entirely. A 2015 FDA approval of an ingestible sensor — embedded in the pill itself and paired with a wearable patch — was built to close exactly that gap by confirming ingestion, not just container access.

Testing for the Drug Itself

Blood, urine, or hair can be tested for the drug or a related marker, giving direct pharmacological evidence that a dose was actually taken in rather than just obtained.

Hair testing has a distinctive advantage: drugs accumulate along the hair shaft over time, so different segments of a single sample can reveal a longer-term dosing history. In research settings, a marker substance such as riboflavin (vitamin B2) is sometimes added to a study pill so its presence in urine can confirm recent ingestion.

Strengths and Limits

Strengths

Electronic monitors capture precise timing without patient recall; drug and biomarker tests give direct pharmacologic confirmation.

Limits

Cost and burden generally confine these methods to smaller studies, and most blood or urine levels reflect only very recent dosing.

Individual differences in drug metabolism also complicate interpreting a blood or urine level, since the same dose can produce different concentrations in different people.

HARDER CASES, 1 OF 3

Measuring Primary Adherence

Recall primary nonadherence from earlier: a new prescription that is never filled at all. It creates a genuine measurement puzzle, because pharmacy dispensing data — the workhorse of most large studies — only exists once a fill happens.

A prescription that was never filled leaves no dispensing record to find. Detecting it requires linking the original prescribing or e-prescribing record to the pharmacy fill record — and many databases capture only one side of that link, leaving primary nonadherence invisible by default.

HARDER CASES, 2 OF 3

Medications That Aren't Pills

Insulin, inhalers, transdermal patches, and topical creams or gels do not fit neatly into “pill count” or “days-supply” logic built around one tablet, once a day.

Insulin

Dose varies day to day with blood sugar, so there is no fixed “one dose a day” to count against.

Inhalers

Counting actuations does not confirm correct technique, which matters as much as the count itself.

Patches

Applied rather than swallowed, so days-supply has to be approximated from package size.

Creams and Gels

Amount used per application is rarely uniform, making dose tracking especially rough.

HARDER CASES, 3 OF 3

Seven Methods, One Spectrum

Every method covered so far sits somewhere on a spectrum between cheap-and-subjective and expensive-and-objective.

Low Cost

Self-report questionnaires. Cheapest and fastest; most exposed to recall and social-desirability bias.

Moderate Cost

Pharmacy dispensing data and pill counts are relatively objective proxies for possession, not ingestion. Diaries remain a form of prospective self-report, carrying the same reporting risks as asking a patient directly.

High Cost

Electronic monitoring, drug concentration testing. Most direct evidence, least practical at large scale.

No method wins outright — the right choice depends on the study's size, budget, and what question it is actually asking. Because every method has different weaknesses, many studies combine two or more to cross-check their findings.

From Measurement to Analysis

Once adherence has been measured, it gets put to work in two major places: helping interpret clinical trial results, and enabling comparative effectiveness research built on real-world data.

In both settings, a central question recurs: was a disappointing result because the drug does not work, or because it was not actually taken?

The Cholestyramine Trial

The Lipid Research Clinics Coronary Primary Prevention Trial tested cholestyramine, a cholesterol-lowering drug taken as a powder mixed into a drink several times a day — a genuinely unpleasant regimen to keep up.

Average adherence came out lower in the cholestyramine group (about 50.8%) than in the placebo group (about 67.3%), largely because of side effects from the drug itself. The lesson: when a treatment is poorly tolerated, adherence problems can blunt its measured effectiveness before the drug's actual pharmacology ever gets a fair test — a theme that runs through the rest of this section.

Per-Protocol and As-Treated Analyses

Because adherence varies, trials sometimes report analyses beyond the primary intention-to-treat comparison: a per-protocol analysis keeps only participants who actually followed the assigned regimen, and an as-treated analysis groups people by what they actually took rather than what they were assigned.

Both can be useful for exploring a drug's effect under good adherence — but both also break randomization's protection against confounding, reopening the door to the healthy-adherer bias seen earlier. They are suggestive, not a substitute for the trial's main result.

How Studies Handle It

Intention-to-treat remains the standard primary analysis. Trials can also plan around adherence directly — a larger sample size to absorb expected dilution, or a run-in period that screens out early nonadherers before randomization. Observational comparative-effectiveness studies often treat adherence as a time-varying exposure, sometimes with marginal structural models built for exactly this kind of shifting exposure.

Adherence in Comparative Effectiveness Research

In real-world comparative effectiveness research built on observational data, adherence is rarely complete, and it can play three different roles in a single study: an outcome in its own right, part of how the exposure itself gets defined, or a confounder that needs active management.

In practice, that often means treating adherence as a time-varying exposure rather than a fixed one — for example, censoring a patient's follow-up once they stop taking the drug (an as-treated approach), or modeling adherence continuously over the study period with methods like marginal structural models.

Interval Selection

The time window chosen for measuring adherence is not a neutral detail — it shapes the answer. The right interval depends on the drug's own pharmacokinetics and on how granular the measurement method is: a short half-life drug can demand a short window, while a dispensing-based measure often needs a longer window than electronic monitoring to say anything meaningful.

Antiretroviral therapy for HIV shows the stakes concretely: a 90-day adherence measure has been more strongly associated with viral load than a 30-day measure of the same behavior. Oral contraceptives cut the other way — because just two consecutive missed pills can meaningfully raise the risk of pregnancy, a short window can matter more than a long one.

CHOOSING INTERVAL AND METRIC, 2 OF 3

The 80% Threshold Debate

Many studies dichotomize adherence into “adherent” versus “nonadherent” at 80%, purely as a matter of convention rather than settled evidence for any particular drug.

Collapsing a continuous measure into a single cut point discards real information and can bias effect estimates, since patients at 79% and 40% get treated identically to each other, and as different from an 81% patient as they are from each other.

The chapter's own guidance: when the dose-response relationship looks roughly linear, adherence is often better analyzed as a continuous measure; when neither a simple binary nor a continuous scale fits well, ordinal categories can be the better compromise.

Multiple Medication Regimens

Chronic disease management often means several drugs at once, which raises a question with no settled answer: should adherence to the whole regimen be averaged across drugs, weighted by clinical importance, or kept separate for each one?

Drug interactions and regimen-level complexity add further texture that a single combined number can flatten out. No consensus method exists, so researchers need to make this choice deliberately and justify it, rather than defaulting to whatever is easiest to compute.

One common shortcut: classify a patient as adherent if they are adherent to at least one drug in the regimen. It is easy to compute, but it can misclassify patients who are keeping up with some components of their treatment while quietly dropping others.

ADHERENCE OVER TIME, 1 OF 3

Same 50%, Two Different Patients

Picture two patients, each with exactly 50% average adherence over a year. One alternates week by week — a fully adherent week, then a fully nonadherent week, back and forth. The other takes it perfectly for six months, then stops completely for six months.

A single summary average treats these two patients as identical. Their health consequences — and the interventions that would actually help each of them — are likely very different.

ADHERENCE OVER TIME, 2 OF 3

Trajectory Modeling

Group-based trajectory modeling is a statistical technique built for exactly this problem. Instead of reducing each patient to one number, it identifies distinct clusters of adherence patterns over time.

Typical patterns include consistently high adherence, gradual decline, early discontinuation, and late improvement. Grouping patients this way gives a richer picture than an average, and can point to which at-risk subgroups actually need an intervention.

ADHERENCE OVER TIME, 3 OF 3

The Franklin Statin Study

Franklin and colleagues applied trajectory modeling to statin users and identified roughly six distinct adherence patterns rather than one average.

What Makes a Trajectory Group Credible

At least about 5% of the cohort in the group; a reasonably narrow confidence interval around its pattern; and a posterior probability of at least about 0.7 that each assigned patient truly belongs there.

Those criteria keep the method from manufacturing more patterns than the data can actually support. Even so, the groups are general archetypes: no single patient follows the fitted curve exactly, and the final number of groups still depends partly on the investigator's judgment, not on the data alone.

Can We Predict Who Will Struggle?

If poor adherence could be predicted early, interventions could be targeted before a patient falls too far behind. Early refill and dispensing patterns — adherence in just the first few months on a drug — turn out to be a genuinely useful early-warning signal for later adherence.

A more humbling finding: in several studies, providers' own clinical predictions of which patients would or would not be adherent were no better than chance. Even more sophisticated tools fare only modestly better — large clinical and demographic models, and newer machine-learning approaches, still achieve just modest accuracy, likely because many of the real drivers of nonadherence never make it into the data they are built from.

Personalized Thresholds

Recall that no single adherence percentage works as a universal cutoff. One emerging response is to stop trying to find one: instead of a fixed threshold for everyone, build a threshold tailored to each patient's own clinical risk.

Lo-Ciganic and colleagues developed models that estimate individualized adherence thresholds associated with outcomes, rather than assuming an 80%-style rule applies equally to a low-risk patient and a high-risk one.

Emerging Technology

Two kinds of new technology are expanding what is measurable and what is fixable. Sensor technology — ingestible sensors, smart pill bottles, wearables — is narrowing the gap between container access and confirmed ingestion.

Communication technology — text-message reminders, smartphone apps, two-way patient-provider messaging — is expanding options for intervening in real time, not just measuring after the fact. Together, they point toward adherence support that responds as behavior happens, rather than being reconstructed months later.

WRAP-UP, 1 OF 2

The Throughline

Adherence is not one yes-or-no behavior. It is three phases — initiation, implementation, persistence — and every method for measuring it is an imperfect proxy carrying its own particular bias, from self-report's social desirability to dispensing data's possession-not-ingestion gap.

How adherence is measured, and over what interval, genuinely changes what a study concludes — in a randomized trial and in a real-world database study alike. And the field is moving away from a single summary number, toward richer trajectories and thresholds built around the individual patient.

WRAP-UP, 2 OF 2

Thank You

Medication adherence sits at an unusual crossroads: it is a deeply human behavior, shaped by cost, belief, forgetfulness, and side effects, yet it is also something researchers must measure with rigor, because it quietly determines whether a proven drug helps anyone at all.

No single measurement method gets the full picture right. Choosing well — and being honest about what was missed — is what turns an adherence number into something a study can actually trust.

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